AI and Skill Premium

A Simple Extension of Acemoglu and Restrepo(2018b)

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Roadmap

The starting point: Acemoglu and Restrepo (2018b)

Core idea

Technology affects labor demand through a race between two forces:

Automation

Machines take over tasks previously performed by workers.

Automation $\Rightarrow$ displacement effect

New tasks

Technology creates new labor-intensive activities.

Task creation $\Rightarrow$ reinstatement effect

This research extends that framework to heterogeneous skills and learning effect.

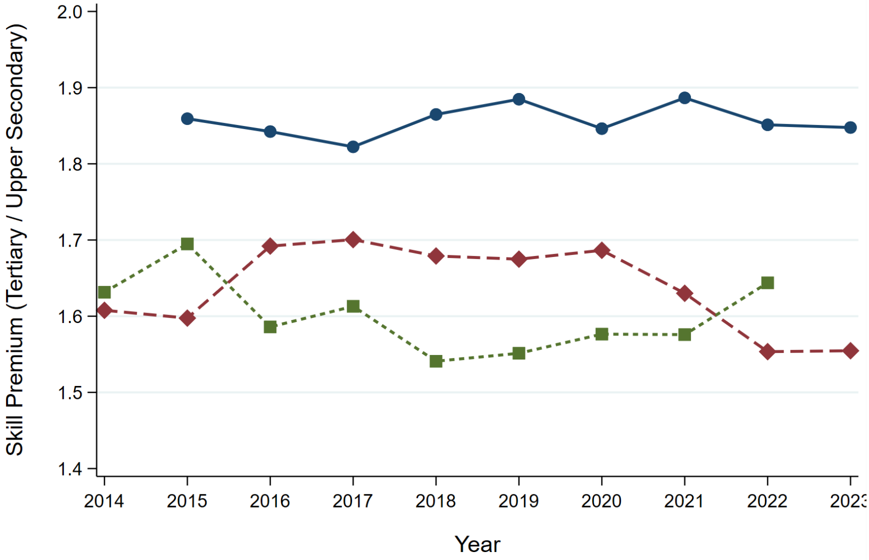
Why extend the model?

- AI is not only a machine that replaces workers; it also creates new tasks, new services, and new production routines.
- High-skill workers may benefit first because they adapt faster to complex AI-enabled tasks.
- Low-skill workers are not passive: through training and production experience, they may gradually learn to perform some new tasks.

Main question

How does AI affect the skill premium when automation, task creation, and worker learning interact over time?

Empirical motivation: bounded skill premia



A simple interpretation of the stylized facts

1. **Persistent level differences.** The United States has a higher skill premium than Germany and the OECD average.
2. **Bounded trajectories.** Even with rapid AI and robot diffusion, wage inequality does not rise without limit.

Interpretation

Institutional and educational systems may shape how quickly low-skill workers learn, retrain, and move into AI-complementary tasks.

Model overview

Three sectors

- Final-good producers
- Intermediate-good producers
- AI research sector

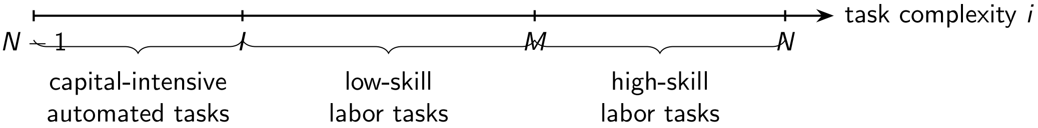
Three inputs

- Capital / machines
- Low-skill labor
- High-skill labor

Key extension

Low-skill productivity is endogenous because workers learn from AI-enabled production.

Production tasks and three thresholds



- I: automation frontier; tasks below I can be done by capital.
- M: skill-allocation threshold; tasks above M require high-skill labor.
- N: frontier of intermediate goods; a higher N means new, more complex tasks.

Intermediate production

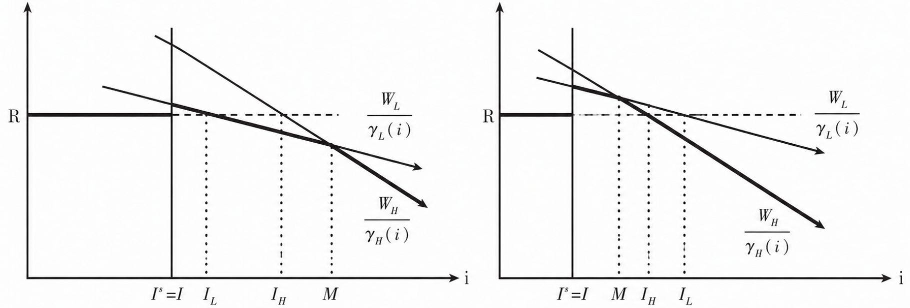
For each task i, output is produced using an AI-related technological product $q(i,t)$ and the cheapest available traditional input:

$$y(i,t) = \begin{cases} q(i,t)^{\eta} \left[k(i,t) + \gamma_L(i,t)l(i,t) + \gamma_H(i,t)h(i,t)\right]^{1-\eta}, & i \leq I(t), \\ q(i,t)^{\eta} \left[\gamma_L(i,t)l(i,t) + \gamma_H(i,t)h(i,t)\right]^{1-\eta}, & i > I(t). \end{cases}$$

Cost-minimizing allocation

Firms choose capital, low-skill labor, or high-skill labor according to the lowest effective unit cost.

Factor choice across tasks



(a) Low-skill labor as the main determinant of investment

(b) High-skill labor as the main determinant of investment

The learning-by-doing mechanism

High-skill labor has a stronger comparative advantage in complex tasks:

$$\gamma_H(i,t) = e^{A_H i}, \qquad A_H > A_L.$$

Low-skill labor can improve through learning:

$$\gamma_L(i,t) = \Gamma(t) e^{A_L i}.$$

The learning term is

$$\Gamma(t) = \beta E(t) l_T(t)^{\alpha} l_P(t)^{1-\alpha}, \qquad l_T(t) + l_P(t) = l_L(t).$$

Interpretation

β measures learning capacity; $E(t)$ measures technological familiarity; l_T and l_P are training and production time.

Skill premium equation

The skill premium is pinned down at the threshold $M(t)$ where firms are indifferent between low- and high-skill labor:

$$\frac{W_L(t)}{\gamma_L(M(t),t)} = \frac{W_H(t)}{\gamma_H(M(t),t)}.$$

Using the productivity functions,

$$\frac{W_H(t)}{W_L(t)} = e^{(A_H - A_L)M(t)}\Gamma(t).$$

Economic meaning

The skill premium rises when the task threshold M moves toward more complex tasks, but it falls when low-skill learning improves relative productivity.

Short-run versus long-run effects

Short run

Learning is weak or fixed.

$$\frac{\partial w}{\partial N} > 0, \qquad \frac{\partial w}{\partial I} > 0.$$

Both new complex tasks and automation raise the skill premium.

Long run

Low-skill workers accumulate experience.

$$\frac{\partial w}{\partial N} < 0, \qquad \frac{\partial w}{\partial I} > 0.$$

Task creation can reduce the skill premium; automation still raises it.

Endogenous AI progress: scientists and research directions

The extension endogenizes AI progress by introducing an AI research sector.

There is a fixed mass of scientists:

$$S > 0.$$

Each scientist chooses between two research directions:

$$S_I(t) + S_N(t) \leq S.$$

Job-replacing research

Scientists develop AI technologies that expand the automation frontier:

$$I(t) \uparrow .$$

These innovations allow capital to replace labor in existing tasks.

Job-creating research

Scientists develop AI technologies that create new frontier tasks:

$$N(t) \uparrow .$$

These innovations expand the range of labor-intensive tasks.

Endogenous AI progress: laws of motion

If scientists work on job-replacing innovation, the automation frontier evolves as

$$\dot{I}(t) = \kappa_I \phi_I(n(t)) S_I(t).$$

If scientists work on job-creating innovation, the task frontier evolves as

$$\dot{N}(t) = \kappa_N \phi_N(n(t)) S_N(t).$$

Here,

$$n(t) \equiv N(t) - I(t)$$

is the measure of labor-intensive tasks.

Interpretation

- κ_I and κ_N measure research productivity.
- $\phi_I(n)$ and $\phi_N(n)$ capture stage-dependent success probabilities.
- When labor-intensive tasks are abundant, automation opportunities are larger.
- When automation is already extensive, creating new human-complementary tasks becomes relatively more valuable.

Endogenous AI progress: how scientists choose

A scientist compares the expected research payoff from the two directions.

Let $V_I(t)$ be the value of a successful job-replacing innovation, and $V_N(t)$ the value of a successful job-creating innovation. Then the research wages are

$$W_I^S(t) = \kappa_I \phi_I(n(t)) V_I(t),$$
$$W_N^S(t) = \kappa_N \phi_N(n(t)) V_N(t).$$

Scientists are heterogeneous in research costs. Scientist j chooses job-replacing research if

$$\frac{W_I^S(t) - W_N^S(t)}{Y(t)} > \chi_I^j - \chi_N^j.$$

If $\chi_I^j - \chi_N^j$ follows a distribution $G(\cdot)$, then

$$S_I(t) = SG\left(\frac{\kappa_I \phi_I(n(t)) V_I(t)}{Y(t)} - \frac{\kappa_N \phi_N(n(t)) V_N(t)}{Y(t)}\right),$$
$$S_N(t) = S - S_I(t).$$

Endogenous AI progress: balanced research allocation

Define normalized innovation values:

$$v_I(n) = \lim_{t\to\infty} \frac{V_I(t)}{Y(t)}, \qquad v_N(n) = \lim_{t\to\infty} \frac{V_N(t)}{Y(t)}.$$

Then the dynamics of labor-intensive tasks are determined by

$$\dot{n}(t) = \dot{N}(t) - \dot{I}(t).$$

Substituting the scientist allocation rule gives

$$\dot{n}(t) = \kappa_N \phi_N(n) S - \left[\kappa_I \phi_I(n) + \kappa_N \phi_N(n)\right] G\left(\kappa_I \phi_I(n) v_I(n) - \kappa_N \phi_N(n) v_N(n)\right) S.$$

On the balanced growth path,

$$\dot{n}(t) = 0,$$

so expected research returns are equalized:

$$\kappa_I \phi_I(n^*) v_I(n^*) = \kappa_N \phi_N(n^*) v_N(n^*).$$

Calibration logic

The quantitative exercise is not designed to estimate the full model structurally. It provides a transparent numerical illustration.

Baseline targets

- Total labor force normalized to 100,000.
- High-skill share: $\omega = 27.05\%$.
- Benchmark occupational skill premium: $w = 1.91$.
- Learning capacity β is calibrated to match the benchmark skill premium.

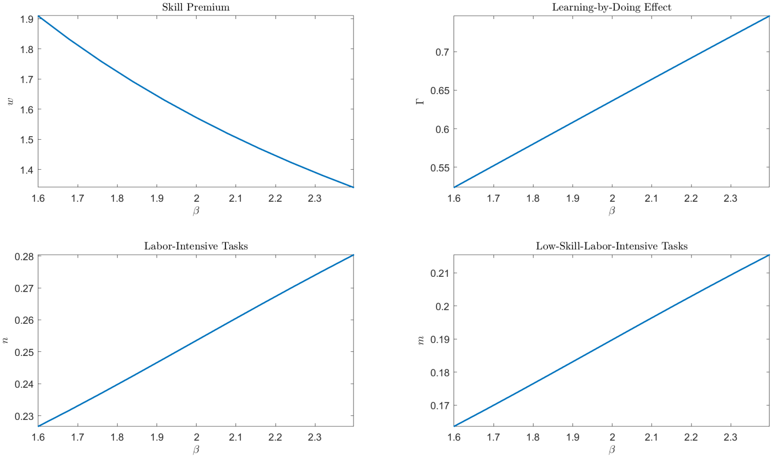
Baseline parameter values

Parameter	$L+H$	S	ω	A_H	A_L	I_H	β
Value	100000	7.17	27.05%	18.68	16.90	26.60%	1.5991
Parameter	α	θ	ρ	σ	η	κ_I	κ_N
Value	0.32	0.80	0.05	0.90	0.86%	12.85%	16.75%

Baseline equilibrium

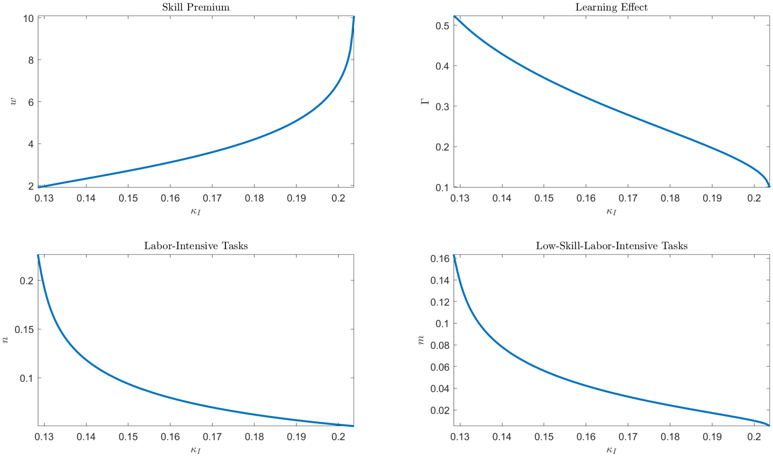
The model matches $w^{model} = 1.91$, with $\Gamma = 0.5236$, $n = 0.2266$, and $m = 0.1635$.

Simulation 1: stronger low-skill learning capacity



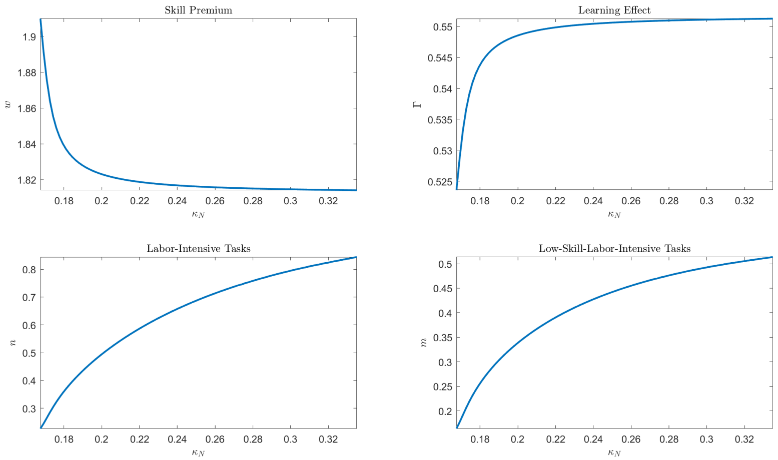
Increasing β by 50% reduces the skill premium from 1.91 to 1.34 and expands low-skill-labor-intensive tasks.

Simulation 2: more productive job-replacing innovation



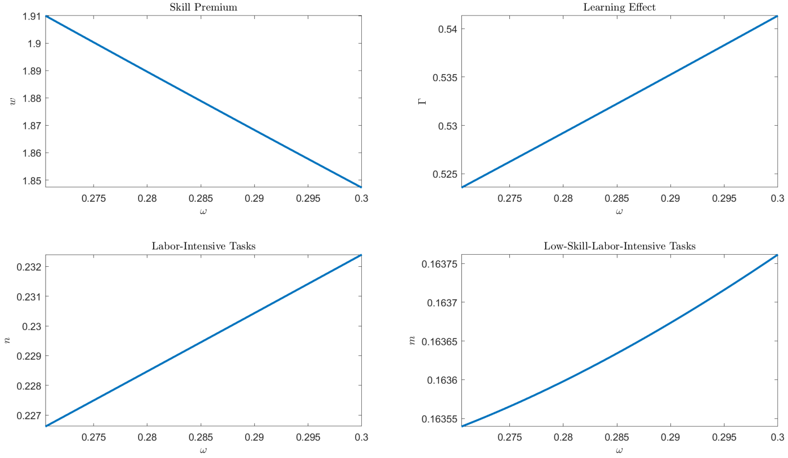
Doubling κ_I modestly reduces the skill premium in the calibrated equilibrium because the induced task reallocation expands learning opportunities.

Simulation 3: more productive job-creating innovation



Doubling κ_N generates a larger expansion of labor-intensive tasks and reduces the skill premium more strongly than doubling κ_I.

Simulation 4: larger high-skill labor share



Increasing the high-skill share from 27.05% to 30% lowers the skill premium from 1.91 to 1.85.

What do the simulations teach us?

1. **Learning capacity matters most.** Improving β has the largest equalizing effect.
2. **Task creation is more inclusive than pure replacement.** Higher κ_N creates more labor-intensive task space.
3. **Education supply helps, but moderately.** Raising ω reduces skill scarcity, but it does not directly transform low-skill task access.

Main message

AI-driven growth is distributionally benign only when workers can move into new tasks rather than being trapped outside them.

Contribution relative to Acemoglu and Restrepo (2018b)

Original task framework	Extension in this presentation
Automation displaces labor	Automation affects low- and high-skill workers differently
New tasks reinstate labor demand	New tasks are first high-skill intensive, then partly absorbed by low-skill workers
Labor productivity is largely fixed after task shocks	Low-skill productivity evolves through learning-by-doing
Growth and factor shares are central	Skill premium becomes the main distributional object

Policy implications

1. Training policy

Vocational education, continuing education, and workplace AI training raise low-skill learning capacity.

2. Innovation policy

Support AI applications that create human-complementary tasks, not only automation tools that replace labor.

3. Education policy

Expanding higher education reduces skill scarcity, but it should be combined with retraining for existing workers.

Limitations and extensions

- The model abstracts from unemployment, search frictions, and firm heterogeneity.
- Institutions such as collective bargaining and employment protection are discussed but not fully modeled.
- Calibration uses aggregate and occupational data; worker-level AI exposure would improve identification.

Natural next step

Add labor-market frictions and heterogeneous firms to study worker transitions across tasks and occupations.

Takeaway

AI does not mechanically raise or lower the skill premium.

Its distributional effect depends on the balance between

automation, **task creation**, and **worker learning**.

The long-run policy challenge is to make task creation accessible to workers who are initially displaced.

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